7	**(a)**	**1.**	The existence of a threshold (minimum) frequency below which no photoelectrons are emitted no matter how intense the EM radiation is.	**1**
		2.	Above the threshold frequency, the maximum kinetic energy of the photoelectron (stopping potential) increases with the frequency and is independent of the intensity.	**1**
		3.	There is no appreciable time delay between the incident EM radiation and the emission of photoelectrons even for very low EM intensities.	**1**

	(b)	**(i)**	Work function energy refers to the minimum energy required to liberate an electron (escape) from the metal surface.	**1**
		(ii)	Photon energy = $\frac{hc}{\lambda}$ $= \frac{(6.63\times10^{-34})(3.00\times10^{8})}{380\times10^{-9}}$ $= 5.23 \times 10^{-19}$ = 3.27 eV	**1** **1**
		(iii)	Sodium and calcium (both must be listed) (allow ecf) The incident photon has energy greater than their work function energies	**1** **1**
	(c)	From de Broglie's equation, $p = \frac{h}{\lambda}$ $p = \frac{6.63\times10^{-34}}{380\times10^{-9}}$ $= 1.74\times10^{-27}$ N s From Newton's 2nd law $F = \frac{\Delta p_{total}}{\Delta t} = \frac{N}{t}\Delta p$ (rate of photons incident on metal)(change in momentum for one photon) $= (7.6 \times 10^{14})(1.74\times10^{-27})$ $= 1.3 \times10^{-12}$ N		**1** **1** **1**

8	(a)		radians	1